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CO4_AT1_DATA INTERPRETATION

1. Real-Time Detection System

The surveillance system has three methods:

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

The requirement is **strict latency constraints while maintaining acceptable accuracy**.

Comparison

Viola-Jones has the highest speed, making it suitable for systems where response time is extremely important. However, its accuracy is only **78%**, which may be insufficient for reliable surveillance.

YOLO provides **92% accuracy** with moderate speed. It offers a good compromise between detection accuracy and processing speed.

Deep Learning provides the highest accuracy of **95%**, but it is slow and therefore may not satisfy strict latency requirements.

Selection

YOLO should be selected.

Justification

YOLO provides significantly better accuracy than Viola-Jones while remaining faster than the Deep Learning method.

- Viola-Jones → 78% accuracy, fast
- YOLO → **92% accuracy, moderate speed**
- Deep Learning → 95% accuracy, slow

For a real-time surveillance system, a small loss in accuracy compared with Deep Learning can be acceptable if it provides substantially better response time.

Deployment Conditions

If the surveillance system operates on limited hardware or requires immediate detection, YOLO is more practical. If latency is absolutely critical and some reduction in accuracy is acceptable, Viola-Jones can be considered. If latency is less important and maximum accuracy is required, Deep Learning would be preferable.

Final Answer

because it provides **92% accuracy with moderate processing speed**, giving a practical balance between accuracy and latency.

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2. Robust Object Detection

The performance under different conditions is:

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Normal Conditions

Under normal conditions:

- Viola-Jones = **80%**
- YOLO = **90%**
- Deep Learning = **95%**

Deep Learning performs best.

Occlusion Conditions

When objects are partially blocked:

- Viola-Jones = **60%**
- YOLO = **75%**
- Deep Learning = **85%**

Deep Learning again provides the highest accuracy.

Low-Light Conditions

Under low-light conditions:

- Viola-Jones = **55%**
- YOLO = **70%**
- Deep Learning = **80%**

Deep Learning has the best performance.

Overall Average

For Viola-Jones:

For YOLO:

For Deep Learning:

Comparison

Method	**Average Performance**
Viola-Jones	65%
YOLO	78.33%
Deep Learning	**86.67%**

Selection

is the best overall method.

Trade-offs

Deep Learning: Highest robustness and accuracy, but it generally requires more computational resources and may have higher latency.

YOLO: Provides a good balance between robustness and speed, making it suitable for real-time applications.

Viola-Jones: Fast and computationally simpler, but its performance decreases considerably under occlusion and low-light conditions.

Final Answer

Deep Learning provides the **best overall performance**, particularly in difficult conditions. However, YOLO may be selected when real-time performance and computational efficiency are more important than maximum accuracy.

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3. Optical Flow Analysis

The two methods are:

- **Method A:** Produces stable motion vectors but misses small motions.
- **Method B:** Detects fine motion but produces noisy vectors.

Comparison

Factor	Method A	Method B
Motion accuracy	Good for large motion	Good for fine motion
Stability	**High**	Lower
Small motion detection	Poor	**Excellent**
Noise sensitivity	Low	**High**
Dynamic scenes	More stable	More detailed but noisy

Method A

Method A is suitable when stable and reliable motion vectors are more important than detecting very small movements. It is less sensitive to noise and can provide consistent tracking.

Method B

Method B can identify small and fine movements that Method A may miss. However, its motion vectors contain more noise, which can result in unstable tracking.

Selection

For **dynamic scenes**, **Method A is generally more suitable when stable tracking is the primary requirement**.

The reason is that dynamic scenes can already contain significant movement, camera motion, and changing objects. Noisy vectors can make tracking unreliable.

However, if the application specifically requires detection of **small or fine movements**, Method B would be preferred, provided that noise-reduction techniques are applied.

Final Answer

Method A provides better stability and lower noise sensitivity. Method B is preferable for applications where detecting small motion is more important than vector stability.

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4. Model Generalization and Overfitting

The given models are:

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

Model A

Training accuracy:

Testing accuracy:

Difference:

The large **28% gap** indicates that Model A has a high risk of **overfitting**.

The model performs extremely well on training data but performs considerably worse on unseen testing data.

Generalization

Model A has **poor generalization ability**.

Model B

Training accuracy:

Testing accuracy:

Difference:

The small **2% gap** indicates that Model B generalizes much better.

The training and testing accuracies are close, indicating that the model has learned patterns that are useful for unseen data.

Comparison

Factor	Model A	Model B
Training Accuracy	98%	90%
Testing Accuracy	70%	**88%**
Accuracy Gap	28%	**2%**
Overfitting Risk	**High**	Low
Generalization	Poor	**Good**

Factor	Model A	Model B
Deployment Reliability	Low	**High**

Selection

should be selected for real-world deployment.

Justification

Although Model A has higher training accuracy, its testing accuracy is only 70%. This means it does not generalize well to unseen data.

Model B achieves 88% testing accuracy and has only a 2% difference between training and testing accuracy. Therefore, it is more reliable for real-world use.

Final Answer

Model B is the more reliable model because it has better generalization, a much smaller training-testing gap, and a lower risk of overfitting.

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5. System Performance Under Constraints

The models are:

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

The objective is to select the model with the **best balance between accuracy, latency and cost** for a real-time application.

Model A

- Accuracy = 88%
- Latency = 50 ms
- Cost = Low

Model A provides the fastest response and lowest cost. However, its accuracy is the lowest.

Model B

- Accuracy = 92%

- Latency = 80 ms
- Cost = Moderate

Model B provides better accuracy than A while maintaining relatively low latency and moderate cost.

Model C

- Accuracy = 95%
- Latency = 120 ms
- Cost = High

Model C has the highest accuracy but also the highest latency and cost.

Comparison

Model	Accuracy	Latency	Cost	Overall
A	88%	**50 ms**	**Low**	Fast and economical
B	**92%**	80 ms	Moderate	**Best balance**
C	**95%**	120 ms	High	Highest accuracy but expensive

Selection

Justification

Model B offers a strong compromise:

- Accuracy increases from 88% to **92%** compared with Model A.
- Latency remains relatively low at **80 ms**.
- Cost is only moderate.
- It avoids the high cost and 120 ms latency of Model C.

For a real-time system, Model B provides a better overall balance than choosing the highest-accuracy model alone.